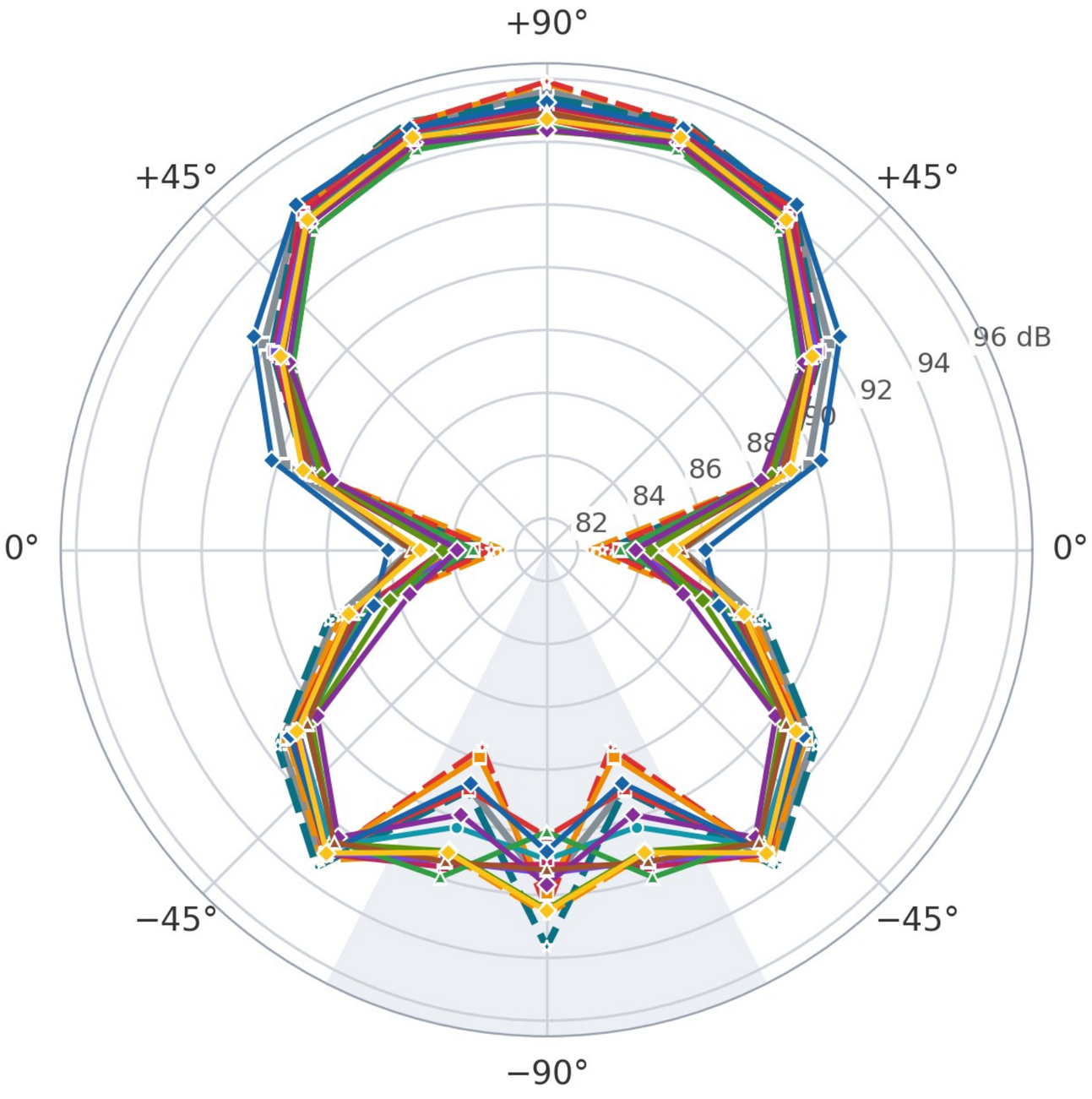


THE DUCT LININGS

Duct linings at 2000 µs

Mean level over the arc. Markers: ★ 3 Nov, ■ 2 Nov, ● 30 Oct, ◆ 29 Oct, ▲ 28 Oct. † lining fitted the old way (before 30 Oct). The bare prop is off the plot.

		level	thrust	rpm
Bare prop, no duct	■ 2 Nov	80.0 dB	**10.56 N**	11,807
Felt, double + rubber †	◆ 29 Oct	90.5 dB	**8.69 N**	12,177
Thinsulate, 4 layers	■ 2 Nov	90.6 dB	**9.22 N**	12,157
Felt †	▲ 28 Oct	90.7 dB	**8.96 N**	12,112
Inlet + Thinsulate, 4 layers	★ 3 Nov	90.9 dB	**10.05 N**	12,076
Thinsulate, 2 layers	■ 2 Nov	90.9 dB	**8.78 N**	12,205
Felt, double layer †	◆ 29 Oct	90.9 dB	**8.62 N**	12,196
Thinsulate, 6 layers	● 30 Oct	91.0 dB	**8.64 N**	12,176
Thinsulate, 10 layers	■ 2 Nov	91.0 dB	**8.89 N**	12,177
PU foam †	▲ 28 Oct	91.1 dB	**8.16 N**	12,234
Thinsulate, 8 layers	■ 2 Nov	91.2 dB	**8.82 N**	12,200
Inlet + Thinsulate, 2 layers	★ 3 Nov	91.3 dB	**10.04 N**	12,087
Rubber †	◆ 29 Oct	91.3 dB	**7.58 N**	12,292
PU foam + rubber †	◆ 29 Oct	91.3 dB	**8.33 N**	12,217
Duct + inlet, no lining	★ 3 Nov	91.4 dB	**10.12 N**	12,078
Duct, no lining	■ 2 Nov	91.5 dB	**7.99 N**	12,269

+1.23 N

Thinsulate, 4 layers against the unlined duct, both 2 Nov: it wins back 48 % of the 2.57 N the empty duct loses to the bare prop, and is 0.9 dB quieter from −54° to +90°.

Every duct is louder than the bare prop: 90.5–91.5 dB against 80.0 dB, at 10.56 N and 11,807 rpm bare. Set-ups marked † are from other days and the old fit: indicative only.

Inlet (dashed, 3 Nov): 10.12 N, +2.13 N over the unlined duct of 2 Nov at the same level (−0.3 dB from −54° to +90°). Same day with the inlet: Thinsulate 2 −0.08 N, −0.2 dB; Thinsulate 4 −0.07 N, −0.3 dB. A repeat of the bare inlet agreed to 0.04 N.

28 Oct–3 Nov 2026, APC 8x4.5MR, AXI 2217/12, 500 Hz–24 kHz, 2000 µs, mean of 3 repeats, corrected files. Axis 81–96.5 dB. Grey: −90°/−72°, probably in the flow. rpm = stand × 60/7.